



Prateek Gupta

Roll No.:EC23B016

Electronics and Communication Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	National Institute of Technology Arunachal Pradesh	8.96 (Current)	2023-27
Senior Secondary	Tathagat Gyansthli Senior Secondary School	87.0%	2022

EXPERIENCE

Research Intern

May 2026 - July 2026

Indian Institute of Technology, Gandhinagar

Gandhinagar, India

- Studied and analyzed Analog In-Memory Computing (AIMC) architecture, including VTDC, Wordline DAC, and SRAM (6T/8T/10T) cells, gaining a strong understanding of analog memory computing and SRAM operation.
- Designed and integrated the full-custom layout of a VTDC with Wordline DAC and a 64×128 SRAM array; performed DRC, LVS, post-layout simulations, and compared pre and post layout performance.

Circuit Development Engineer Intern

Jun 2025 - Aug 2025

Ornate Solar Pvt. Ltd.

New Delhi, India

- Designed and simulated a Proportional-Integral (PI) Controller to regulate the output voltage of a DC-DC converter, ensuring stable voltage under varying input and load conditions.
- Implemented the feedback control algorithm using Python-based discrete-time simulation and developed an analog behavioral model in Verilog-A, validating controller performance through behavioral analysis.

Project Assistance Intern

Oct 2025 - Dec 2025

I-CAS Lab NIT Arunachal Pradesh

Jote, India

- Learned and worked on Cadence design tools under a Dynamic Power Management project.
- Practiced analog flow full-custom layout design, gaining an understanding of **DRC** and **LVS** verifications.

PROJECTS AND CONFERENCES

A Design Framework of Dynamic Stress Control for NBTI-Resilient Power Gating

July 2026

IEEE International Symposium on Very Large Scale Integration (ISVLSI) 2026 [Tapout SCL180nm]

Vijay Pratap Yadav, Prateek Gupta, Vipin Kumar Singh, Vassilis Alimisis, Paul P.Sotiriadis, Sanjeev Kumar Metya

- Engineered a circuit to mitigate Negative Bias Temperature Instability [NBTI] effects by dynamically switching transistors, thereby improving the lifetime of the power-gating circuitry.
- Completed the full-chip layout including floorplanning, device matching, routing, and parasitic-aware optimization for tapeout readiness.

Arithmetic and Logical unit | Verilog HDL,

Feb 2025 – Mar 2025

- Operation Selection: User can choose from various option like Addition, Subtraction, Multiplication, Division, Equality checker, Inequality checker with the help of operator input.
- There are two operands each of 8 bits, User can enter data as per instructions mention in ALU Testbench (instructions may vary for different operation).
- Two output each of 8 bits, output will be shown as per operations.

Comparator (Inverter Based) | Cadence Virtuoso, Layout

July 2025 – Sept 2025

- Designed a NOT gate-based comparator that outputs a digital 0 or 1 depending on whether the input signal exceeds a predefined threshold voltage.
- Completed the full custom Layout in SCL180nm CMOS technology.

TECHNICAL SKILLS

Developer Skills: VLSI IC Designing, Custom Layout Designing, Verilog Hdl, C/C++, Python

Designing Tools: Cadence Virtuoso and Layout, Xilinx Vivado, Icarus Verilog, Gtkwave, Ltspice, VS Code

Course Work: Memory Devices, Digital Integrated Circuit, Network Theory, MOSFET, Semiconductor Devices, Digital Electronics, Signal and System, Analog Electronics, COA

CERTIFICATION COURSES

[.75] **India Space Lab:** Summer Technical Training Program

[.75] **CeNSE Winter School on semiconductor Technology:** Certificate of Completion